

FAILURE CRITERIA FOR DUCTILE MATERIALS

Main general and particular cases

GENERAL CASE

$\sigma_A, \sigma_B, \sigma_C$ principal stresses:

Tresca: $\sigma_{eq} = \max (| \sigma_A - \sigma_B |, | \sigma_B - \sigma_C |, | \sigma_C - \sigma_A |)$

Von Mises: $\sigma_{eq} = (1/\sqrt{2}) \sqrt{(\sigma_A - \sigma_B)^2 + (\sigma_B - \sigma_C)^2 + (\sigma_C - \sigma_A)^2}$

GENERAL CASE

$\sigma_{xx}, \sigma_y, \sigma_z, \tau_{xy}, \tau_{yz}, \tau_{xz}$:

Tresca: σ_{eq} as before, it needs computing the 3 principal stress components

Von Mises: $\sigma_{eq} = (1/\sqrt{2}) \sqrt{(\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + 6(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{xz}^2)}$

CASE OF ORDERED PRINCIPAL STRESSES

$\sigma_1 \geq \sigma_2 \geq \sigma_3$ principal stresses:

Tresca: $\sigma_{eq} = \sigma_1 - \sigma_3$

Von Mises: $\sigma_{eq} = (1/\sqrt{2}) \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}$

CASE OF PLANE STRESS STATE, KNOWN PRINCIPAL STRESSES

σ_A & σ_B , principal stresses, $\sigma_C = 0$ (biaxial state of stress):

Tresca: $\sigma_{eq} = \max (| \sigma_A - \sigma_B |, | \sigma_A |, | \sigma_B |)$

Von Mises: $\sigma_{eq} = \sqrt{\sigma_A^2 + \sigma_B^2 - \sigma_A \sigma_B}$

CASE OF SIMPLE STRESS STATE σ - τ , xy REFERENCE

$\sigma_{yy} = \sigma_{zz} = \tau_{xz} = \tau_{yz} = 0$:

Tresca: $\sigma_{eq} = \sqrt{\sigma_{xx}^2 + 4 \tau_{xy}^2}$

Von Mises: $\sigma_{eq} = \sqrt{\sigma_{xx}^2 + 3 \tau_{xy}^2}$

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Other particular cases

CASE OF PLANE STRESS STATE, xy REFERENCE

$$\sigma_{yy} = \sigma_{zz} = \tau_{xz} = \tau_{yz} = 0:$$

Tresca: $\sigma_{eq} = \max(\sqrt{(\sigma_{xx}^2 + \sigma_{yy}^2) + 4 \tau_{xy}^2}, \sqrt{(\sigma_{xx}^2 + 4 \tau_{xy}^2)}, \sqrt{(\sigma_{yy}^2 + 4 \tau_{xy}^2)})$

Von Mises: $\sigma_{eq} = \sqrt{(\sigma_{xx}^2 + \sigma_{yy}^2) - \sigma_{xx}\sigma_{yy} + 3 \tau_{xy}^2}$

CASE OF UNIAXIAL STRESS STATE

$$\sigma_{yy} = \sigma_{zz} = \tau_{xy} = \tau_{yz} = \tau_{xz} = 0:$$

Tresca: $\sigma_{eq} = \sigma_{xx}$

Von Mises: $\sigma_{eq} = \sigma_{xx}$

CASE OF PURE SHEAR

$$\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \tau_{xz} = \tau_{yz} = 0:$$

Tresca: $\sigma_{eq} = 2 \tau_{xy}$

Von Mises: $\sigma_{eq} = \sqrt{3} \tau_{xy}$

FAILURE CRITERIA FOR BRITTLE MATERIALS

Main general and particular cases

GENERAL CASE

$\sigma_A, \sigma_B, \sigma_C$ principal stresses:

Rankine: $\sigma_{eq} = \max(\sigma_A, \sigma_B, \sigma_C)$

CASE OF ORDERED PRINCIPAL STRESSES

$\sigma_1 \geq \sigma_2 \geq \sigma_3$ principal stresses:

Rankine: $\sigma_{eq} = \sigma_1$

CASE OF PLANE STRESS STATE, KNOWN PRINCIPAL STRESSES

σ_A & σ_B , principal stresses, $\sigma_C = 0$ (biaxial state of stress):

Rankine: $\sigma_{eq} = \max(\sigma_A, \sigma_B, 0)$

CASE OF SIMPLE STRESS STATE σ - τ , xy REFERENCE

$\sigma_{yy} = \sigma_{zz} = \tau_{xz} = \tau_{yz} = 0$:

Rankine: $\sigma_{eq} = \max(\frac{1}{2} \sigma_{xx} \pm \frac{1}{2} \sqrt{[\sigma_{xx}^2 + 4 \tau_{xy}^2]})$

CASE OF UNIAXIAL STRESS STATE

$\sigma_{yy} = \sigma_{zz} = \tau_{xy} = \tau_{yz} = \tau_{xz} = 0$:

Rankine: $\sigma_{eq} = \sigma_{xx}$

CASE OF PURE SHEAR

$\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \tau_{xz} = \tau_{yz} = 0$:

Rankine: $\sigma_{eq} = \tau_{xy}$